```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import numpy as np                                   # Implemennts milti-dimensional array and matr
import pandas as pd                                  # For data manipulation and analysis
# import pandas_profiling
import matplotlib.pyplot as plt                      # Plotting library for Python programming lang
import seaborn as sns                                # Provides a high level interface for drawing 
%matplotlib inline
sns.set()

from subprocess import check_output ###Assignment
```

```
# loading of the data
Tesla_df = pd.read_csv("/content/drive/MyDrive/Abhijit/Capstone/Tesla - Deaths.csv")

Tesla_df.head()
```

	Case #	Year	Date	Country	State	Description	Deaths	Tesla driver	Tesla occupant	Other vehicle	...	Verified Tesla Autopilot Deaths	Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO	
0	294	2022	1/17/23	USA	CA	Tesla crashes into back of semi	1	1	-	-	...	-	-	http
1	293	2022	1/7/23	Canada	-	Tesla crashes	1	1	-	-	...	-	-	http
2	292	2022	1/7/23	USA	WA	Tesla hits pole, catches on fire	1	-	1	-	...	-	-	http
3	291	2022	12/22/22	USA	GA	Tesla crashes and burns	1	1	-	-	...	-	-	http
4	290	2022	12/19/22	Canada	-	Tesla crashes into storefront	1	-	-	-	...	-	-	http

5 rows × 24 columns

1. Preliminary data inspection and cleaning

```
Tesla_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 294 entries, 0 to 293
Data columns (total 24 columns):
 #   Column                                                              Non-Null Count  Dtype
---  ------                                                              --------------  -----
 0   Case #                                                              294 non-null    int64
 1   Year                                                                294 non-null    int64
 2   Date                                                                294 non-null    object
 3    Country                                                            294 non-null    object
 4    State                                                              294 non-null    object
 5    Description                                                        294 non-null    object
 6    Deaths                                                             294 non-null    int64
 7    Tesla driver                                                       289 non-null    object
 8    Tesla occupant                                                     285 non-null    object
 9    Other vehicle                                                      290 non-null    object
 10   Cyclists/ Peds                                                     291 non-null    object
 11   TSLA+cycl / peds                                                   292 non-null    object
 12   Model                                                              294 non-null    object
 13   Autopilot claimed                                                  276 non-null    object
 14   Verified Tesla Autopilot Deaths                                    290 non-null    object
 15   Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO  293 non-null    object
 16  Unnamed: 16                                                         289 non-null    object
 17  Unnamed: 17                                                         289 non-null    object
 18   Source                                                             294 non-null    object
 19   Note                                                               9 non-null      object
```

```
 20   Deceased 1                                               87 non-null     object
 21   Deceased 2                                               17 non-null     object
 22   Deceased 3                                               4 non-null      object
 23   Deceased 4                                               0 non-null      float64
dtypes: float64(1), int64(3), object(20)
memory usage: 55.3+ KB
```

```
Tesla_df.shape
```

```
(294, 24)
```

```
Tesla_df.columns
```

```
Index(['Case #', 'Year', 'Date', ' Country ', ' State ', ' Description ',
       ' Deaths ', ' Tesla driver ', ' Tesla occupant ', ' Other vehicle ',
       ' Cyclists/ Peds ', ' TSLA+cycl / peds ', ' Model ',
       ' Autopilot claimed ', ' Verified Tesla Autopilot Deaths ',
       ' Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO ',
       'Unnamed: 16', 'Unnamed: 17', ' Source ', ' Note ', ' Deceased 1 ',
       ' Deceased 2 ', ' Deceased 3 ', ' Deceased 4 '],
      dtype='object')
```

- From the data information it is clear that the some data has some missing values I don't need some of the columns like 'unnamed 16', 'unnamed 17', 'note' and 'source' so we can remove them also the deceased counts are very less with respect to the entries of the accidents. Therefore, these columns are not at all useful for analysis. Basically we can remove them.

```
# Data cleaning by column removal
Tesla_df_clean = Tesla_df.drop(['Unnamed: 16', 'Unnamed: 17', ' Source ', ' Note ', ' Deceased 1 ',
       ' Deceased 2 ', ' Deceased 3 ', ' Deceased 4 '], axis=1)
```

```
Tesla_df_clean.head()
```

	Case #	Year	Date	Country	State	Description	Deaths	Tesla driver	Tesla occupant	Other vehicle	Cyclists/ Peds	TSLA+cycl / peds	Model
0	294	2022	1/17/23	USA	CA	Tesla crashes into back of semi	1	1	-	-	-	1	-
1	293	2022	1/7/23	Canada	-	Tesla crashes	1	1	-	-	-	1	-
2	292	2022	1/7/23	USA	WA	Tesla hits pole, catches on fire	1	-	1	-	-	1	-

Next steps: Generate code with Tesla_df_clean New interactive sheet

```
Tesla_df_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 294 entries, 0 to 293
Data columns (total 16 columns):
 #   Column                                                   Non-Null Count  Dtype
---  ------                                                   --------------  -----
 0   Case #                                                   294 non-null    int64
 1   Year                                                     294 non-null    int64
 2   Date                                                     294 non-null    object
 3    Country                                                 294 non-null    object
 4    State                                                   294 non-null    object
 5    Description                                             294 non-null    object
 6    Deaths                                                  294 non-null    int64
 7    Tesla driver                                            289 non-null    object
 8    Tesla occupant                                          285 non-null    object
 9    Other vehicle                                           290 non-null    object
 10   Cyclists/ Peds                                          291 non-null    object
 11   TSLA+cycl / peds                                        292 non-null    object
 12   Model                                                   294 non-null    object
 13   Autopilot claimed                                       276 non-null    object
 14   Verified Tesla Autopilot Deaths                         290 non-null    object
 15   Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO  293 non-null    object
dtypes: int64(3), object(13)
memory usage: 36.9+ KB
```

```
Tesla_df_clean.columns
```

```
Index(['Case #', 'Year', 'Date', ' Country ', ' State ', ' Description ',
       ' Deaths ', ' Tesla driver ', ' Tesla occupant ', ' Other vehicle ',
       ' Cyclists/ Peds ', ' TSLA+cycl / peds ', ' Model ',
       ' Autopilot claimed ', ' Verified Tesla Autopilot Deaths ',
       ' Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO '],
      dtype='object')
```

```
Tesla_df_clean.columns = Tesla_df_clean.columns.str.strip() # To strip the space in the ccolumn names
```

```
Tesla_df_clean.columns
```

```
Index(['Case #', 'Year', 'Date', 'Country', 'State', 'Description', 'Deaths',
       'Tesla driver', 'Tesla occupant', 'Other vehicle', 'Cyclists/ Peds',
       'TSLA+cycl / peds', 'Model', 'Autopilot claimed',
       'Verified Tesla Autopilot Deaths',
       'Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO'],
      dtype='object')
```

```
# Check of the records where tesla occupants were present
Tesla_df_clean['Tesla occupant'].value_counts()
```

Tesla occupant	count
-	242
1	39
2	3
3	1

dtype: int64

```
Tesla_df_clean['Tesla driver'].unique()
```

```
array(['1', ' - ', nan], dtype=object)
```

```
# Checking of the missing value percentage in the cleaned data
Tesla_df_clean.isnull().sum()/len(Tesla_df_clean)*100
```

	0
Case #	0.000000
Year	0.000000
Date	0.000000
Country	0.000000
State	0.000000
Description	0.000000
Deaths	0.000000
Tesla driver	1.700680
Tesla occupant	3.061224
Other vehicle	1.360544
Cyclists/ Peds	1.020408
TSLA+cycl / peds	0.680272
Model	0.000000
Autopilot claimed	6.122449
Verified Tesla Autopilot Deaths	1.360544
Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO	0.340136

dtype: float64

```
Tesla_df_clean['Autopilot claimed'].unique()
```

```
array([' - ', '1', nan, '2'], dtype=object)
```

```
Tesla_df_clean.fillna(' - ', inplace=True)
```

```
Tesla_df_clean['Autopilot claimed'].unique()
```

```
array([' - ', '1', '2'], dtype=object)
```

```
# Checking of the missing value percentage after replacing the nan with " - " the cleaned data
Tesla_df_clean.isnull().sum()/len(Tesla_df_clean)*100
```

	0
Case #	0.0
Year	0.0
Date	0.0
Country	0.0
State	0.0
Description	0.0
Deaths	0.0
Tesla driver	0.0
Tesla occupant	0.0
Other vehicle	0.0
Cyclists/ Peds	0.0
TSLA+cycl / peds	0.0
Model	0.0
Autopilot claimed	0.0
Verified Tesla Autopilot Deaths	0.0
Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO	0.0

dtype: float64

```
Tesla_df_clean.head()
```

	Case #	Year	Date	Country	State	Description	Deaths	Tesla driver	Tesla occupant	Other vehicle	Cyclists/ Peds	TSLA+cycl / peds	Model
0	294	2022	1/17/23	USA	CA	Tesla crashes into back of semi	1	1	-	-	-	1	-
1	293	2022	1/7/23	Canada	-	Tesla crashes	1	1	-	-	-	1	-
2	292	2022	1/7/23	USA	WA	Tesla hits pole, catches on fire	1	-	1	-	-	1	-

Next steps: Generate code with Tesla_df_clean New interactive sheet

```
Tesla_df_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 294 entries, 0 to 293
Data columns (total 16 columns):
 #   Column                                                             Non-Null Count  Dtype 
---  ------                                                             --------------  ----- 
 0   Case #                                                             294 non-null    int64 
 1   Year                                                               294 non-null    int64 
 2   Date                                                               294 non-null    object
 3   Country                                                            294 non-null    object
 4   State                                                              294 non-null    object
 5   Description                                                        294 non-null    object
 6   Deaths                                                             294 non-null    int64 
 7   Tesla driver                                                       294 non-null    object
 8   Tesla occupant                                                     294 non-null    object
 9   Other vehicle                                                      294 non-null    object
 10  Cyclists/ Peds                                                     294 non-null    object
 11  TSLA+cycl / peds                                                   294 non-null    object
 12  Model                                                              294 non-null    object
 13  Autopilot claimed                                                  294 non-null    object
 14  Verified Tesla Autopilot Deaths                                    294 non-null    object
 15  Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO  294 non-null    object
```

```
dtypes: int64(3), object(13)
memory usage: 36.9+ KB
```

```
Tesla_df_clean['Date'] = pd.to_datetime(Tesla_df_clean['Date'], format='%m/%d/%y', errors='coerce') # pandas dat
```

```
# dropping year coloum as Date column already has year
Tesla_df_clean = Tesla_df_clean.drop(['Year'], axis=1)
```

```
Tesla_df_clean.head()
```

	Case #	Date	Country	State	Description	Deaths	Tesla driver	Tesla occupant	Other vehicle	Cyclists/ Peds	TSLA+cycl / peds	Model	Autopilo claime
0	294	2023-01-17	USA	CA	Tesla crashes into back of semi	1	1	-	-	-	1	-	
1	293	2023-01-07	Canada	-	Tesla crashes	1	1	-	-	-	1	-	
2	292	2023-	USA	WA	Tesla hits pole,	1		1			1		

Next steps: Generate code with Tesla_df_clean New interactive sheet

2. Exploratory Data Analysis

```
# year wise accident analysis
Tesla_df_clean['Date'].dt.year.value_counts()
```

Date	count
2022	93
2021	58
2019	46
2020	39
2018	18
2016	15
2017	11
2015	5
2014	4
2023	3
2013	2

dtype: int64

Maximum accident has occured dureing with years sugegsts possibly use of tesla car increased so accidents are also increased

```
# How many crashes happen over time?
events_per_date = Tesla_df_clean.groupby('Date').size()
events_per_date
```

Date	0
2013-04-02	1
2013-11-02	1
2014-07-04	2
2014-07-14	1
2014-12-30	1
...	...
2022-12-18	1
2022-12-19	1
2022-12-22	1
2023-01-07	2
2023-01-17	1

263 rows × 1 columns

dtype: int64

```
# Monthwise Analysis of Accident
Tesla_df_clean['Date'].dt.month.value_counts()
```

Date	count
11	35
12	33
5	28
6	28
7	27
9	26
8	24
3	24
4	23
1	21
10	16
2	9

dtype: int64

In the month of May, june, july, Nov, and Dec the number of accident records are high

```
# Date wise Analysis of accident
Tesla_df_clean['Date'].dt.day.value_counts()
```

	count
Date	
7	21
17	13
18	13
1	13
12	13
21	13
20	13
26	12
24	12
11	12
4	12
8	12
22	11
14	11
10	11
13	10
28	9
19	9
15	9
16	8
29	8
23	7
5	7
30	6
2	5
25	5
6	4
3	4
27	4
31	4
9	3

dtype: int64

```
# Date wise Analysis of death having high number of Death records
Tesla_df_clean['Date'].dt.day.value_counts()[Tesla_df_clean['Date'].dt.day.value_counts() > 10]
```

	count
Date	
7	21
17	13
18	13
1	13
12	13
21	13
20	13
26	12
24	12
11	12
4	12
8	12
22	11
14	11
10	11

dtype: int64

There is no specific trend of accident cases with the date

```
# Weekwise analysis of death
Tesla_df_clean['Date'].dt.weekday.value_counts()
```

	count
Date	
5	53
6	49
4	45
0	43
2	41
1	35
3	28

dtype: int64

Accidents rates are very near the weekends having indices 5,6 over the weekdays having indices 0,1,2,3. Therefore possibly more peoples were driving car in weekends

```
Tesla_df_clean['Country'].value_counts()
```

	count
Country	
USA	215
China	16
Germany	11
Canada	10
Netherlands	6
UK	5
Norway	4
Holland	3
Switzerland	3
Taiwan	3
Japan	2
Belgium	2
Denmark	2
Australia	2
France	2
Finland	1
Mexico	1
South Korea	1
Portugal	1
Austria	1
Slovenia	1
Spain	1
Ukraine	1

dtype: int64

USA has siginificant records of the accidents. Possibly they are the heavy user of this car.

```
# Death percentages in USA
Tesla_df_clean[Tesla_df_clean['Country'].str.strip() == 'USA']['Deaths'].sum()/Tesla_df_clean['Deaths'].sum()*10
```

```
np.float64(73.93767705382436)
```

```
Tesla_df_clean['Country'].unique()
```

```
array([' USA ', ' Canada ', ' China ', ' Mexico ', ' UK ', ' Germany ',
       ' Finland ', ' Australia ', ' Netherlands ', ' Switzerland ',
       ' France ', ' Denmark ', ' Belgium ', ' Portugal ',
       ' South Korea ', ' Norway ', ' Taiwan ', ' Slovenia ', ' Austria ',
       ' Ukraine ', ' Spain ', ' Holland ', ' Japan '], dtype=object)
```

```
Tesla_df_clean[Tesla_df_clean['Country'].str.strip() == 'USA']['State'].value_counts()
```

	count
State	
CA	75
FL	24
CA	16
AZ	8
FL	8
OH	6
GA	6
NY	5
PA	5
NV	4
VA	4
UT	4
TX	4
MO	3
NJ	3
GA	3
SC	2
CO	2
NH	2
HI	2
IL	2
IN	2
IA	2
MI	2
OR	2
PA	2
WA	2
MD	1
NY	1
IL	1
MO	1
WA	1
AR	1
HA	1
UT	1
ME	1
AL	1
MA	1
NC	1
ID	1
DE	1
TN	1

dtype: int64

```
Tesla_df_clean['Deaths'].sum()
```

```
np.int64(353)
```

```
Tesla_df_clean[Tesla_df_clean['State'].str.strip() == 'CA']['Deaths'].sum()/Tesla_df_clean['Deaths'].sum()*100
```

```
np.float64(31.1614730878187)
```

31% of the overall death occured in CA state only

```
Tesla_df_clean.columns
```

```
Index(['Case #', 'Date', 'Country', 'State', 'Description', 'Deaths',
       'Tesla driver', 'Tesla occupant', 'Other vehicle', 'Cyclists/ Peds',
       'TSLA+cycl / peds', 'Model', 'Autopilot claimed',
       'Verified Tesla Autopilot Deaths',
       'Verified Tesla Autopilot Deaths + All Deaths Reported to NHTSA SGO'],
      dtype='object')
```

```
# Checkthe accident involved differnt numbers of deaths
Tesla_df_clean['Deaths'].value_counts()
```

Deaths	count
1	247
2	38
3	6
4	3

dtype: int64

All the accident involed one or many death cases. Fortunately, in most of the accidents the number of death cases was 1. a few cases observed where the number of deaths are more than 2.

```
# Check the accident involved differnt numbers of drivers
Tesla_df_clean['Tesla driver'].value_counts()/len(Tesla_df_clean)*100
```

Tesla driver	count
-	60.204082
1	39.795918

dtype: float64

In 60% accident cases the drivers were allived suggests somehow car safety is moderatley good

```
Tesla_df_clean['Tesla occupant'].value_counts()
```

Tesla occupant	count
-	251
1	39
2	3
3	1

dtype: int64

There are 251 cases where no Tesla occupants were dead. 43 tesla occupants were dead according to the report

```
# Check the accident involved different numbers of drivers where a Tesla driver was involved
Tesla_df_clean[Tesla_df_clean['Tesla driver'] == '1']['Tesla occupant'].value_counts()
```

Tesla occupant	count
-	96
1	18
2	2
3	1

dtype: int64

there are 18 cases where one Occupant was dead, 2 cases where 2 occupants were dead and 1 cases where 3 occupants were dead along with the driver

```
Tesla_df_clean['Other vehicle'].value_counts().sum()
```

```
np.int64(294)
```

```
Tesla_df_clean['Other vehicle'].value_counts()/Tesla_df_clean['Other vehicle'].value_counts().sum()*100
```

Other vehicle	count
-	62.585034
1	32.312925
2	3.741497
3	1.020408
4	0.340136

dtype: float64

So, 62% of the accident does not involves hit to any other vehicle but rest of the percentages (32%) are involved with hit of some other cars possibly failure in the detection/mechanical techniques

```
Tesla_df_clean['Cyclists/ Peds'].value_counts()/Tesla_df_clean['Cyclists/ Peds'].value_counts().sum()*100
```

Cyclists/ Peds	count
-	85.034014
1	14.285714
2	0.680272

dtype: float64

Around 15% times Tesla car hit the pedestrian/cyclist. Possibly engine failure or vission detection problem.

```
(pd.to_numeric(Tesla_df_clean['Tesla driver'], errors='coerce').sum()/Tesla_df_clean['Deaths'].sum())*100
```

```
np.float64(33.14447592067989)
```

```
(pd.to_numeric(Tesla_df_clean['Tesla occupant'], errors='coerce').sum()/Tesla_df_clean['Deaths'].sum())*100
```

```
np.float64(13.59773371104816)
```

33% was driver from the total death cases, 14% was the occupants Therefore, rests were either pedestrians or cyclist (~53%). It is a very terrific report about the automous car

```
tesla_occupant_numeric = pd.to_numeric(Tesla_df_clean['Tesla occupant'], errors='coerce').fillna(0)
cyclists_peds_numeric = pd.to_numeric(Tesla_df_clean['Cyclists/ Peds'], errors='coerce').fillna(0)
tesla_driver_numeric = pd.to_numeric(Tesla_df_clean['Tesla driver'], errors='coerce').fillna(0)

count = Tesla_df_clean[
    (tesla_occupant_numeric > 0) &
    (cyclists_peds_numeric > 0) &
```

```
    (tesla_driver_numeric == 1)
].shape[0]

print(count)
```

```
0
```

There are no such accidents where driver, tesla occupant, and cyclist/pedestrians all were involved

```
Tesla_df_clean['Model'].value_counts()
```

Model	count
-	181
S	45
3	39
X	17
Y	10
1	1
2	1

dtype: int64

Among All the accidents record, S model has highest accident rates follwed by 3 and X. Probably, S,3, X have some safety issues.

```
Tesla_df_clean['Model'].unique()
```

```
array([' - ', ' Y ', '1', '2', '3', ' S ', ' X '], dtype=object)
```

```
Tesla_df_clean[Tesla_df_clean['Model']==' S ']['Deaths'].value_counts()
```

Deaths	count
1	38
2	6
3	1

dtype: int64

```
Tesla_df_clean[Tesla_df_clean['Model']==' S ']['Cyclists/ Peds'].value_counts()
```

Cyclists/ Peds	count
-	40
1	5

dtype: int64

```
Tesla_df_clean[Tesla_df_clean['Model']=='3']['Deaths'].value_counts()
```

Deaths	count
1	33
2	4
3	2